Japan Technology: Semiconductor Capital Equipment: Micron technology webinar read-across: Benefits for TEL/Disco/Ebara/Kokusai

Shuhei Nakamura
+81(3)4587-9932 |
shuhei.nakamura@gs.com
Goldman Sachs Japan Co., Ltd.

Kaho Otake
+81(3)4587-7498 | kaho.otake@gs.com
Goldman Sachs Japan Co., Ltd.

Late at night on October 21 (JST), memory major Micron Technology (covered by US Semiconductors analyst Jim Schneider) held a technology and product webinar, providing updates on its technology roadmaps for both DRAM and NAND. Below, we briefly summarize the key points and the implications for Japanese SPE makers.

As the updates relate to technology roadmaps, the actual timing of mass production will still depend largely on future demand trends. Nevertheless, we view the expanding application of hybrid bonding and the move toward higher layer counts in NAND as having overall positive implications for **Tokyo Electron (Buy)**, **Disco (Buy)**, **Ebara (Buy)**, and **Kokusai Electric (Neutral)**.

DRAM: Parallel development of scaling and stacking (3D-DRAM) technologies

Micron has already entered mass production for its 1-gamma node using EUV lithography, and management plans to drive further scaling through the adoption of hybrid bonding and high-NA EUV lithography. In parallel, development is ongoing on 3D-DRAM, which stacks memory cells vertically. Micron indicated that it no longer intends to pursue development of $4F^2$ DRAM technology, which vertically stacks transistors to reduce area, from the perspective of manufacturing process complexity and cost. It will instead scale $6F^2$ in the interim.

HBM: Advancing technologies in pursuit of higher bandwidth and lower power consumption

Various developments are underway for HBM, aimed at further increasing bandwidth and reducing power consumption per token. In addition to the current standard HBM architecture, development is progressing on custom HBM, Vertical HBM (vertically stacking HBM dies), and VEIM (Vertical Edge Inference Memory). The company indicated that because all of these architectures require highly sophisticated advanced packaging technologies, it believes technologies such as hybrid bonding and temperature control will become indispensable.

NAND: Transition from current G9 to 400+ layer G11

Following on from generation 9 NAND (276 layers of memory cells), which is currently in mass production, Micron plans to develop its generation 11. G11 will feature more than 400 memory cell layers and utilize hybrid bonding technology for

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

wafer connection, with the company aiming for mass production in 2H CY27. To enable AI-class NAND, development is also underway for SLC (Single Level Cell) based solutions as well as NAND architectures that utilize TSV (Through Silicon Via) technology with a logic base die, enabling a high-capacity tier close to the accelerator.

Implications for Japanese SPE makers

Historically, the mass-production application of hybrid bonding for connecting wafers has been limited to CMOS image sensors and certain 3D-NAND manufactures (such as Kioxia). However, as applications expand across DRAM and a broader range of NAND makers going forward, we believe this will very likely expand business opportunities for **Tokyo Electron's bonders, Disco's (high-clean specification) dicers and grinders,** and **Ebara's CMP systems**.

In addition, we think the move toward higher layer counts in NAND will substantially benefit **Kokusai Electric**, by driving an increase in the number of processes utilizing batch ALD as well as generational shifts in equipment. It should also benefit **Tokyo Electron**, by expanding the TAM for its cryo-etching equipment. Moreover, an expansion in demand for SLC-based NAND would mean that the number of wafers required per bit would increase several times over compared with TLC or QLC (Lam Research noted that a 1% shift toward SLC in NAND bit demand would require an additional 2% in wafer capacity), which we believe would have positive implications for NAND equipment demand.

Exhibit 1: Target prices, valuation methodologies, and risks for mentioned stocks

Company Name (rating)	Ticker	12-m TP (¥)	Methodology	Risks
DISCO (Buy)	6146.T	105,000	Based on the global SPE sector average EV/EBITDA multiple of 18X and our FY3/28E estimates. We then apply a sector-relative premium of 50%.	(-) slowdown or share loss in AI-related demand (-) slowdown in China demand or tightening of export controls (-) rapid yen appreciation against the USD
Ebara (Buy)	6361.T	7,900	Based on the correlation between P/B and our FY12/27E ROE estimates.	(-) Semiconductor capex enters a downcycle (-) Increasing competitiveness of Chinese CMP system makers (-) Slow adoption of new technology in semiconductor devices (-) Decline in crude oil/LNG prices, oil refining/petrochemical margins
Kokusai Electric (Neutral)	6525.T	10,500	Based on the global SPE sector average EV/EBITDA multiple of 18X and our FY3/28E estimates.	(+/-) Changes in investment at key customers (+/-) Additional changes in export controls (+/-) Changes in the competitive landscape due to strategic changes at competitors
Tokyo Electron (Buy)	8035.T	18,000	Based on the global SPE sector average EV/EBITDA multiple of 18X and our FY3/28E estimates. We then apply a sector-relative premium of 30%.	(-) prolonged inventory adjustment phase in the semiconductor industry (-) further strengthening of export restrictions (-) depressed valuation multiple amid upward pressure on interest rates and other factors

Source: Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Shuhei Nakamura and Kaho Otake, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Shuhei Nakamura Goldman Sachs Japan Co., Ltd., Kaho Otake Goldman Sachs Japan Co., Ltd..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

DISCO (Buy, ¥51,230), Ebara (Buy, ¥3,827), Kokusai Electric (Neutral, ¥6,798) and Tokyo Electron (Buy, ¥9,450)

The rating(s) for DISCO, Ebara, Kokusai Electric and Tokyo Electron is/are relative to the other companies in its/their coverage universe: Advantest, DISCO, Ebara, HOYA, JEOL, Kioxia Holdings, Kokusai Electric, Lasertec, SCREEN Holdings, Tokyo Electron, Tokyo Seimitsu, Ulvac

The rating(s) for Micron Technology Inc. is/are relative to the other companies in its/their coverage universe: ARM Holdings, Accenture Plc, Advanced Micro Devices Inc., Amkor Technology Inc., Analog Devices Inc., Applied Materials Inc., Broadcom Inc., Cadence Design Systems Inc., Camtek, Cognizant Technology Solutions, Credo Technology Group, EPAM Systems Inc., Entegris Inc., GlobalFoundries Inc., Globant SA, Intel Corp., International Business Machines Corp., KLA Corp., Lam Research Corp., MKS Instruments Inc., Marvell Technology Inc., Microchip Technology Inc., Micron Technology Inc., NXP Semiconductors NV, Nvidia Corp., ON Semiconductor Corp., Onto Innovation, Qnity, Qualcomm Inc., SanDisk Corp., Seagate Technology, SiTime Corp., Synopsys Inc., TaskUs Inc., Teradyne Inc., Texas Instruments Inc., Western Digital Corp.

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs beneficially owned 1% or more of common equity (excluding positions managed by affiliates and business units not required to be aggregated under US securities law) as of the month end preceding this report: DISCO (¥51,230), Ebara (¥3,827) and Kokusai Electric (¥6,798)

Goldman Sachs has received compensation for investment banking services in the past 12 months: Kokusai Electric (¥6,798) and Micron Technology Inc. ($856.49)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: DISCO (¥51,230), Ebara (¥3,827), Kokusai Electric (¥6,798), Micron Technology Inc. ($856.49) and Tokyo Electron (¥9,450)

Goldman Sachs has received compensation for non-investment banking services during the past 12 months: Tokyo Electron (¥9,450)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: DISCO (¥51,230), Ebara (¥3,827), Kokusai Electric (¥6,798), Micron Technology Inc. ($856.49) and Tokyo Electron (¥9,450)

Goldman Sachs had a non-investment banking securities-related services client relationship during the past 12 months with: Micron Technology Inc. ($856.49) and Tokyo Electron (¥9,450)

Goldman Sachs had a non-securities services client relationship during the past 12 months with: DISCO (¥51,230), Ebara (¥3,827), Micron Technology Inc. ($856.49) and Tokyo Electron (¥9,450)

Goldman Sachs is a registered market maker in the securities or derivatives thereof: Micron Technology Inc. ($856.49)

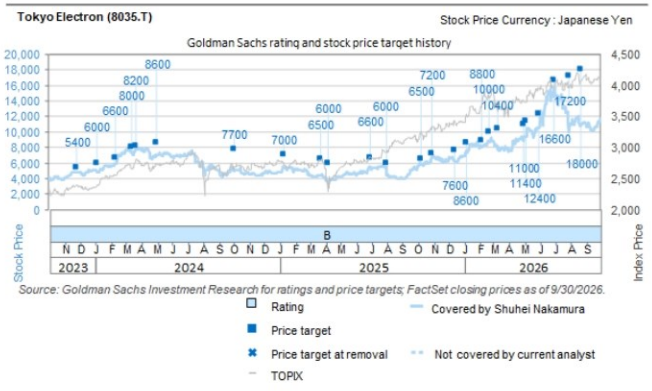
Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

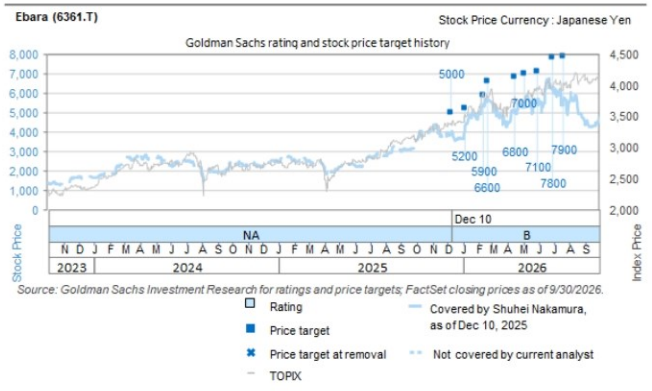
	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	58%	43%

As of October 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,094 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

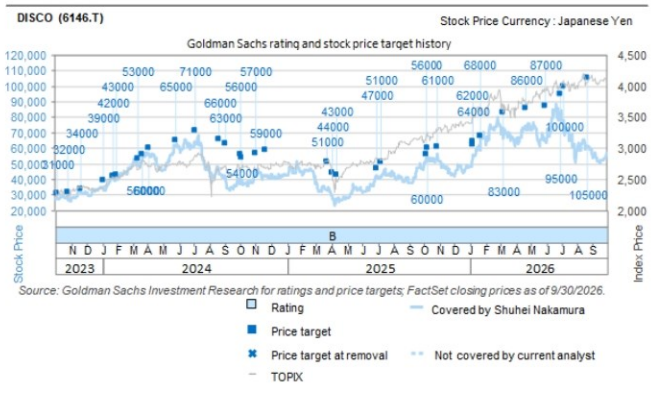
Price target and rating history chart(s)



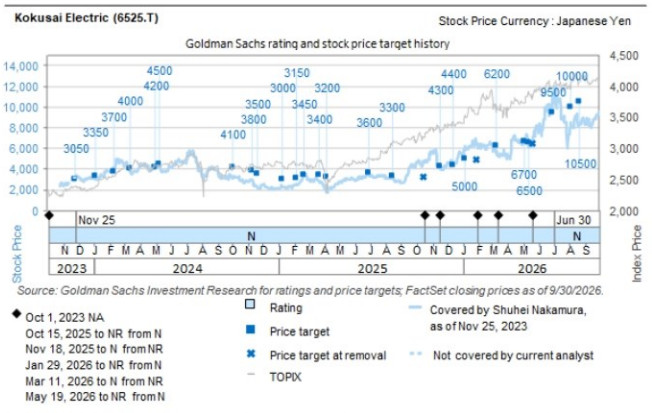
The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



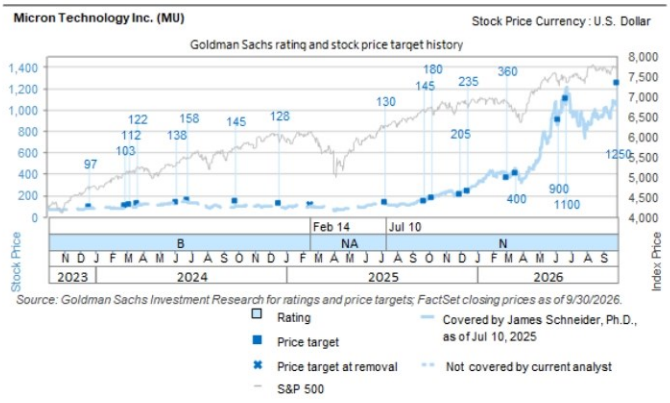
The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

DISCO (6146.T)

Date of report	Target price (¥)	Closing price (¥)
23-Aug-26	105,000	61,360
06-Jul-26	100,000	78,000
30-Jun-26	95,000	81,260
31-May-26	87,000	65,090
22-Apr-26	86,000	74,830
09-Mar-26	83,000	66,500
21-Jan-26	68,000	58,570
08-Jan-26	64,000	55,590
06-Jan-26	62,000	54,200
29-Oct-25	61,000	56,390
08-Oct-25	60,000	52,140
06-Oct-25	56,000	54,000
09-Jul-25	51,000	41,290
30-Jun-25	47,000	42,630
13-Apr-25	43,000	27,470
04-Apr-25	44,000	27,635
24-Mar-25	51,000	33,060
21-Nov-24	59,000	42,380
01-Nov-24	57,000	42,810
04-Oct-24	54,000	39,710
02-Oct-24	56,000	37,410
02-Sep-24	63,000	41,350
22-Aug-24	66,000	43,560
04-Jul-24	71,000	64,580
27-May-24	65,000	61,790
04-Apr-24	60,000	56,750
21-Mar-24	56,000	52,960
13-Mar-24	53,000	50,000
31-Jan-24	43,000	40,380
24-Jan-24	42,000	40,730
04-Jan-24	39,000	33,630
21-Nov-23	34,000	32,030
26-Oct-23	32,000	26,960

Tokyo Electron (8035.T)

Date of report	Target price (¥)	Closing price (¥)
23-Aug-26	90,000	10,858
30-Jul-26	86,000	10,448
30-Jun-26	83,000	15,430
31-May-26	62,000	10,484
04-May-26	57,000	9,490
30-Apr-26	55,000	8,878
09-Mar-26	52,000	7,784
21-Feb-26	50,000	8,792
06-Feb-26	44,000	8,206
06-Jan-26	43,000	7,470
15-Dec-25	38,000	6,228
31-Oct-25	36,000	6,836
08-Oct-25	32,500	5,851
31-Jul-25	30,000	5,466
30-Jun-25	33,000	5,536
07-Apr-25	30,000	3,412
24-Mar-25	32,500	4,438
10-Jan-25	35,000	5,405
02-Oct-24	38,500	5,016
02-May-24	43,000	7,002
21-Mar-24	41,000	7,868
13-Mar-24	40,000	7,478
09-Feb-24	33,000	5,951
04-Jan-24	30,000	4,801
25-Nov-23	27,000	4,801

Ebara (6361.T)

Date of report	Target price (¥)	Closing price (¥)
22-Jul-26	7,900	5,593
30-Jun-26	7,800	6,256
31-May-26	7,100	5,683
04-May-26	7,000	5,242

Kokusai Electric (6525.T)

Date of report	Target price (¥)	Closing price (¥)
23-Aug-26	10,500	8,321
06-Aug-26	10,000	8,224
30-Jun-26	9,500	10,855
13-May-26	6,500	6,170

Ebara (6361.T)

Date of report	Target price (¥)	Closing price (¥)
15-Apr-26	6,800	5,004
21-Feb-26	6,600	5,638
13-Feb-26	5,900	5,303
06-Jan-26	5,200	4,050
10-Dec-25	5,000	3,942

Kokusai Electric (6525.T)

Date of report	Target price (¥)	Closing price (¥)
04-May-26	6,700	6,500
11-Mar-26	6,200	5,768
06-Jan-26	5,000	5,812
15-Dec-25	4,400	4,516
18-Nov-25	4,300	4,033
15-Aug-25	3,300	3,066
30-Jun-25	3,600	3,467
07-Apr-25	3,200	1,790
24-Mar-25	3,400	2,803
21-Feb-25	3,450	3,629
07-Feb-25	3,150	2,518
10-Jan-25	3,000	2,242
21-Nov-24	3,500	2,195
11-Nov-24	3,800	2,953
02-Oct-24	4,100	3,130
10-May-24	4,500	4,275
02-May-24	4,200	4,105
13-Mar-24	4,000	3,770
09-Feb-24	3,700	4,100
04-Jan-24	3,350	2,983
25-Nov-23	3,050	3,220

Micron Technology Inc. (MU)

Date of report	Target price ($)	Closing price ($)
01-Oct-26	1,250.00	1,097.39
25-Jun-26	1,100.00	1,213.56
08-Jun-26	900.00	949.28
19-Mar-26	400.00	444.27
02-Mar-26	360.00	412.67
18-Dec-25	235.00	248.55
03-Dec-25	205.00	234.16
08-Oct-25	180.00	196.54
23-Sep-25	145.00	166.41
10-Jul-25	130.00	123.11
19-Dec-24	128.00	87.09
26-Sep-24	145.00	109.88
27-Jun-24	158.00	132.23
04-Jun-24	138.00	126.64
21-Mar-24	122.00	109.85
07-Mar-24	112.00	98.98
26-Feb-24	103.00	89.46
21-Dec-23	97.00	85.48

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and

regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at https://www.gs.com/research/hedge.html.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that

has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our

research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. You may not use, extract, download or retrieve this information, in whole or in part, to train, finetune, adjust, influence the weights or behavior of, or otherwise develop or improve a machine learning or artificial intelligence model or system, or provide or reproduce this information, in whole or in part, as a prompt or input to any such system without the express written consent of Goldman Sachs. Furthermore, you may not bulk download this information, or employ any robot, spider, crawler, scraper, automated agent, or other electronic means to scrape, harvest, extract, or retrieve this information, without the express written consent of Goldman Sachs. A breach of the above terms may result in termination of your right to access this service.